

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the seven parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Use the atomic masses given in each question. A periodic table and the answer key are at the back.

Parts 1-2 · Equations and ratios

Q1 CR1-LO1

Which is the balanced equation for iron rusting, $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$?

- A. $2\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$
- B. $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
- C. $\text{Fe} + \text{O}_3 \rightarrow \text{FeO}_3$
- D. $2\text{Fe} + 3\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$

Q2 CR1-LO1

Balance $\text{N}_2 + __\text{H}_2 \rightarrow 2\text{NH}_3$. What coefficient goes in front of H_2 ?

Q3 CR1-LO1

It is acceptable to change a subscript in a formula to balance an equation.

True / False

Q4 CR1-LO1

12.0 g of magnesium burns completely in 8.0 g of oxygen to form magnesium oxide only. What mass of magnesium oxide forms?

Q5 CR1-LO2

Using $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$, how many O_2 molecules react with 10 C_2H_2 molecules?

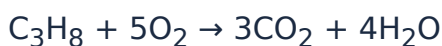
Part 3 · Types of reaction

Q6 CR1-LO3

Classify each reaction.

Options: combination · decomposition · combustion









Parts 4-5 · Redox

Q7 CR1-LO4

In $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, which substance is oxidized?

- A. Zn
- B. Cu^{2+}
- C. Zn^{2+}
- D. Cu

Q8 CR1-LO4

In the pacemaker battery reaction $2\text{Li} + \text{I}_2 \rightarrow 2\text{LiI}$, the reducing agent is:

- A. I_2
- B. Li
- C. LiI
- D. none

Q9 CR1-LO4

Complete the half reaction: $\text{Al} \rightarrow \text{Al}^{3+} + __ \text{e}^-$

Q10 CR1-LO4

Is each change an oxidation or a reduction?

Options: oxidation · reduction

ethanol $\rightarrow$ acetaldehyde (loses 2 H)

pyruvic acid $\rightarrow$ lactic acid (gains 2 H)

$\text{CH}_4 \rightarrow \text{CO}_2$

$\text{CO}_2 \rightarrow$ glucose (photosynthesis)

Q11 CR1-LO4

An antioxidant such as vitamin C works as a reducing agent.

True / False

Part 6 · The mole and molar mass

Q12 CR1-LO5

Calculate the molar mass of CO_2 ($\text{C} = 12.01$, $\text{O} = 16.00$).

Q13 CR1-LO5

How many moles are in 36.04 g of water (18.02 g/mol)?

Q14 CR1-LO5

How many moles of O atoms are in 2 mol of H_2SO_4 ?

- A. 2
- B. 4
- C. 6
- D. 8

Q15 CR1-LO5

How many molecules are in 0.500 mol of glucose?

- A. 0.500
- B. 3.011×10^{23}
- C. 6.022×10^{23}
- D. 1.204×10^{24}

Part 7 • Stoichiometry

Q16 CR1-LO6

Using $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many moles of O_2 are needed to make 5.0 mol of H_2O ?

Q17 CR1-LO6

$\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$. What mass of CO_2 (44.01 g/mol) forms from 10.0 g of CaCO_3 (100.09 g/mol)?

Pharmacy scenario · An antacid tablet

Scenario

A milk of magnesia tablet contains **500 mg of magnesium hydroxide**. It neutralizes stomach acid: $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$. (Mg = 24.31, O = 16.00, H = 1.01, Cl = 35.45)

Q18 CR1-LO5

Calculate the molar mass of $\text{Mg}(\text{OH})_2$.

Q19 CR1-LO5

How many moles of $\text{Mg}(\text{OH})_2$ are in the 500 mg tablet? (Give 3 significant figures.)

Q20 CR1-LO6

How many moles of HCl can the tablet neutralize?

Q21 CR1-LO6

What mass of HCl is that (HCl = 36.46 g/mol)? Explain why the answer is not simply 0.500 g.

Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the seven parts of the lecture and end with a pharmacy scenario.

Answer key

Parts 1-2 · Equations and ratios

Q1 CR1-LO1

Which is the balanced equation for iron rusting, $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$?



4 Fe and 6 O on each side, in the lowest ratio.

Q2 CR1-LO1

Balance $\text{N}_2 + __\text{H}_2 \rightarrow 2\text{NH}_3$. What coefficient goes in front of H_2 ?

3

2NH_3 contains 6 H, so 3H_2 is needed.

Q3 CR1-LO1

It is acceptable to change a subscript in a formula to balance an equation.

True / False → **False**

Changing a subscript changes the substance. Only coefficients change.

Q4 CR1-LO1

12.0 g of magnesium burns completely in 8.0 g of oxygen to form magnesium oxide only. What mass of magnesium oxide forms?

20 g

Conservation of mass: $12.0 + 8.0 = 20.0$ g.

Q5 CR1-LO2

Using $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$, how many O_2 molecules react with 10 C_2H_2 molecules?

25 molecules

$10 \times (5 \text{ O}_2 / 2 \text{ C}_2\text{H}_2) = 25$.

Part 3 · Types of reaction

Q6 CR1-LO3

Classify each reaction.

$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2 \rightarrow$ decomposition; $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O} \rightarrow$ combustion; $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 \rightarrow$ combination; $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow$ decomposition

One $\rightarrow$ many is decomposition; $+ \text{O}_2$ giving oxides is combustion; many $\rightarrow$ one is combination.

Parts 4-5 · Redox

Q7 CR1-LO4

In $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, which substance is oxidized?

A. Zn

B. Cu^{2+}

C. Zn^{2+}

D. Cu

Zn loses two electrons ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$).

Q8 CR1-LO4

In the pacemaker battery reaction $2\text{Li} + \text{I}_2 \rightarrow 2\text{LiI}$, the reducing agent is:

- A. I_2
- B. Li**
- C. LiI
- D. none

Li is oxidized to Li^+ , so it is the reducing agent.

Q9 CR1-LO4

Complete the half reaction: $\text{Al} \rightarrow \text{Al}^{3+} + __e^-$

3

Aluminium loses three electrons; the charge is 0 on each side.

Q10 CR1-LO4

Is each change an oxidation or a reduction?

ethanol $\rightarrow$ acetaldehyde (loses 2 H) $\rightarrow$ oxidation; pyruvic acid $\rightarrow$ lactic acid (gains 2 H) $\rightarrow$ reduction; $\text{CH}_4 \rightarrow \text{CO}_2 \rightarrow$ oxidation; $\text{CO}_2 \rightarrow$ glucose (photosynthesis) $\rightarrow$ reduction

Lose H or gain O = oxidation; gain H or lose O = reduction.

Q11 CR1-LO4

An antioxidant such as vitamin C works as a reducing agent.

True / False $\rightarrow$ **True**

It is oxidized itself, protecting other molecules from oxidation.

Part 6 • The mole and molar mass

Q12 CR1-LO5

Calculate the molar mass of CO_2 (C = 12.01, O = 16.00).

44.01 g/mol

$12.01 + 2 \times 16.00 = 44.01 \text{ g/mol}$.

Q13 CR1-LO5

How many moles are in 36.04 g of water (18.02 g/mol)?

2 mol

$36.04 \text{ g} \div 18.02 \text{ g/mol} = 2.000 \text{ mol}$.

Q14 CR1-LO5

How many moles of O atoms are in 2 mol of H_2SO_4 ?

A. 2

B. 4

C. 6

D. 8

Each H_2SO_4 has 4 O, so $2 \times 4 = 8 \text{ mol O}$.

Q15 CR1-LO5

How many molecules are in 0.500 mol of glucose?

A. 0.500

B. 3.011×10^{23}

C. 6.022×10^{23}

D. 1.204×10^{24}

$0.500 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$.

Part 7 • Stoichiometry

Q16 CR1-LO6

Using $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many moles of O_2 are needed to make 5.0 mol of H_2O ?

2.5 mol

$5.0 \times (1 \text{ mol O}_2 / 2 \text{ mol H}_2\text{O}) = 2.5 \text{ mol}$.

Q17 CR1-LO6

$\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$. What mass of CO_2 (44.01 g/mol) forms from 10.0 g of CaCO_3 (100.09 g/mol)?

4.4 g

$10.0 \div 100.09 = 0.0999 \text{ mol CaCO}_3 \rightarrow 0.0999 \text{ mol CO}_2 (1 : 1) \times 44.01 = 4.40 \text{ g}$.

Pharmacy scenario · An antacid tablet

Q18 CR1-LO5

Calculate the molar mass of $\text{Mg}(\text{OH})_2$.

58.33 g/mol

$24.31 + 2 \times (16.00 + 1.01) = 58.33 \text{ g/mol}$.

Q19 CR1-LO5

How many moles of $\text{Mg}(\text{OH})_2$ are in the 500 mg tablet? (Give 3 significant figures.)

0.00857 mol

$0.500 \text{ g} \div 58.33 \text{ g/mol} = 0.00857 \text{ mol}$.

Q20 CR1-LO6

How many moles of HCl can the tablet neutralize?

0.0171 mol

$0.00857 \text{ mol} \times (2 \text{ mol HCl} / 1 \text{ mol Mg}(\text{OH})_2) = 0.0171 \text{ mol}$.

Q21 CR1-LO6

What mass of HCl is that ($\text{HCl} = 36.46 \text{ g/mol}$)? Explain why the answer is not simply 0.500 g.

$0.0171 \text{ mol} \times 36.46 \text{ g/mol} = \mathbf{0.625 \text{ g HCl}}$. Grams cannot be compared directly, because $\text{Mg}(\text{OH})_2$ and HCl have different molar masses and react in a 1 : 2 **mole** ratio. Always go grams $\rightarrow$ moles $\rightarrow$ mole ratio $\rightarrow$ moles $\rightarrow$ grams.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 5.0–5.6 and 6.0–6.5. Shared under the same licence for non-commercial teaching.